

REV. 3 – 21/02/2025

TRUSTED CARBON REDUCTION (TCR)

METHODOLOGY FOR THE CALCULATION, MONITORING AND VERIFICATION OF GREENHOUSE GAS REDUCTIONS IN AGRICULTURAL ACTIVITIES (GHG)

1. Methodology Description.....	3
1.1 Introduction.....	3
1.2 Methodologies and Standards Used.....	3
2. Objectives.....	4
2.1 Accurate and Transparent Quantification of GHG Emissions and Removals.....	4
2.2 Continuous Monitoring of Environmental Performance in Agriculture.....	5
2.3 Clear and Consistent Reporting of Results.....	5
2.4 Support for the Transition to Sustainable Agricultural Systems.....	5
3. Applicability Conditions.....	6
3.1 Scope of Application.....	6
3.2 Data Availability.....	6
3.3 Resources and Competencies.....	6
3.4 Compliance with Standards.....	7
3.5 Limitations.....	7
4. Methodological Approach.....	7
4.1: System Definition and SSR Identification.....	7
4.2: Baseline Determination.....	8
4.3: Quantification of Agricultural System Emissions and Removals.....	8
4.4: Calculation of Net GHG Reductions/Removals.....	8
4.5: Monitoring and Reporting.....	9
4.6: Conversion of Net GHG Reductions/Removals into Carbon Credits.....	9
4.7: Notarisation of Carbon Credits.....	10
5. Quantification of GHG Emission Mitigations.....	10
5.1: Criteria and Procedures for Quantification.....	10
1. Carbon Removal Relative to the Standardised Baseline.....	10
2. Carbon Removal Relative to Project Carbon-Farming Activities.....	11
3. GHG Increase Relative to the Project Scenario.....	12

4. Leakage.....	13
5. Reserve (Permanence Buffer).....	13
6. Baseline.....	13
6.1 Description.....	13
6.2 Justification.....	13
6.3 Assumptions.....	14
6.4 Lack of Incentives for Change.....	14
6.5 Additionality.....	14
7. GHG Sources, Sinks and Reservoirs.....	14
7.1 Identification of Carbon Reservoirs.....	14
7.2 Identification of Greenhouse Gases.....	15
8. Example of Determination of Sampling and Monitoring Areas.....	16
8.1 Carbon Reserves.....	16
8.2 Carbon Estimates in Different Stocks.....	17
8.3 Delimitation of Project Spatial Boundaries.....	21
8.4 Role of Remote Sensing and Geographic Information Systems (GIS).....	22
8.5 Sampling Design.....	22
8.6 Permanent Sample Plot.....	22
8.7 Calculation of the Number of Permanent Sample Plots.....	23
8.8 Assignment of Number of Sample Units Among Strata.....	24
8.9 Monitoring Plan.....	24
8.10 Reporting.....	24
8.11 Verification.....	25
9. Permanence Risk Assessment.....	25
9.1 Internal Risks.....	25
9.2 External Risks.....	29
9.3 Natural Risks.....	32
9.4 Overall Risk Assessment.....	32
10. Agricultural Practices.....	33
10.1: Recommended Agricultural Practices (BAPs).....	33
10.2: Practices Applied by the Methodology.....	34
10.2.1. Widespread Promotion of Organic Agricultural Management.....	34
10.2.2. Conservation Tillage: Minimum Soil Tillage and Zero Tillage.....	35
10.2.3. Grassland / Use of Cover Crops Year-Round.....	35
10.2.4. Intercropping.....	35
10.2.5. Agricultural Management with Integrated Hedgerows, Rows and Woods in Crops.....	36
10.2.6. Management of Pruning Residues as Carbon Source for SOC.....	36
10.2.7. Application of Natural Inorganic Substances and Natural Fertilisers.....	36
10.2.8. Radical Reduction of Synthetic Fertilisers.....	36
10.2.9. Radical Reduction of Pesticides.....	37
10.2.10. Optimal Organic Matter Recycling.....	37
10.2.11. New Plantations.....	37
11. Parameters, Formulas and Verification.....	38

11.1: Recommended Agricultural Practices (BAPs).....	38
1. Organic Farming (Certified and Non-Certified).....	38
2a. Zero Tillage.....	39
2b. Minimum Soil Tillage.....	39
3a. Green Cover: Spontaneous or Sown Vegetation.....	40
3b. Use of Cover Crops.....	40
4. Intercropping.....	41
5. Integrated Hedgerows, Rows and Woods.....	41
6a. Pruning Residues as Amendment.....	42
6b. Pruning Residues as Energy.....	43
7. Application of Natural Inorganic Substances and Leaf-Based Natural Fertilisers (Rocks or Mineral Dusts).....	43
8. Radical Reduction of Synthetic Fertilisers.....	44
9. Radical Reduction of Pesticides.....	45
10. Optimal Organic Matter Recycling.....	45
11. New Plantations.....	47
12. Glossary of Terms and Acronyms.....	49
Terms.....	49
Acronyms.....	50

1. Methodology Description

1.1 Introduction

This methodology provides a comprehensive and rigorous framework for quantifying, monitoring and reporting greenhouse gas (GHG) emissions and removals in the context of agricultural activities. It was developed to respond to the growing demand for reliable and transparent tools for assessing the environmental impact of agricultural practices and to support the transition towards more sustainable agricultural systems.

The methodology is grounded on a solid scientific basis and integrates components derived from internationally recognised methodologies and standards, adapting them to the specific context of agricultural activities. This approach ensures a high level of credibility, accuracy and comparability of results.

The purpose of the methodology is to provide a valid tool for identifying requirements and procedures for determining project boundaries, identifying the reference scenario, assessing additionality, monitoring relevant parameters and, ultimately, quantifying GHG emission reductions or removals arising from afforestation, reforestation and the adoption of sustainable agricultural practices.

1.2 Methodologies and Standards Used

The main methodological frameworks adopted are as follows:

1. LIFE C-Farms: ([reference to LIFE C-Farms document])

LIFE C-Farms is a European project that developed a plan for reducing GHG emissions in agriculture. The methodology proposed in this document is largely inspired by this framework, adopting its principles and procedures for quantifying emissions and assessing the environmental impact of different practices.

2. Verra VM0042 v2.0: ([reference to Verra VM0042 document])

Verra VM0042 is a Verified Carbon Standard (VCS) for Improved Agricultural Land Management (ALM). This standard provides a robust and standardised approach for quantifying, monitoring and verifying soil carbon sequestration activities, offering a clear methodology for measuring carbon stored in soil following specific management practices. This methodology uses the Verra VM0042 framework to quantify GHG removals associated with soil carbon sequestration practices.

3. CDM's AR-AMS0007 v3.1: ([reference to CDM AR-AMS0007 document])

CDM's AR-AMS0007 is a small-scale afforestation and reforestation (A/R) methodology under the Clean Development Mechanism (CDM) of the United Nations Framework Convention on Climate Change (UNFCCC). This simplified methodology, based on a baseline and monitoring approach, is used to estimate net GHG emission reductions from A/R activities on non-wetland land. This methodology integrates the principles and procedures of AR-AMS0007 for quantifying emissions and removals associated with afforestation/reforestation activities.

4. IPCC Guidelines for National Greenhouse Gas Inventories: ([reference to IPCC guidelines])

The IPCC Guidelines provide a comprehensive and internationally recognised framework for quantifying GHG emissions. This methodology uses the IPCC Guidelines as the basis for estimating emissions from agricultural practices, integrating them with the above-mentioned frameworks to ensure a complete and accurate approach.

The integration of these methodologies and standards creates a synergistic approach that enables a comprehensive and rigorous treatment of the various aspects of quantifying, monitoring and reporting GHG emissions and removals in the context of agricultural activities.

2. Objectives

This document aims to define a system for quantifying, monitoring and reporting greenhouse gas (GHG) emissions and removals in the agricultural sector, with the goal of providing a concrete contribution to the transition towards more sustainable production systems. Its specific objectives include:

2.1 Accurate and Transparent Quantification of GHG Emissions and Removals

A rigorous and transparent framework is established for quantifying GHG emissions and removals related to agricultural activities, achievable through the following:

- **System Definition:** Precisely delineates the boundaries of the agricultural system under analysis, identifying all relevant GHG sources, sinks and reservoirs.
- **Choice of Quantification Methodologies:** Identifies and describes the most appropriate methodologies for estimating GHG emissions and removals, ensuring consistency with international standards and sector best practices.
- **Data Collection:** Defines rigorous processes for collecting accurate and reliable data on the agricultural practices employed, soil properties and other factors influencing the quantification of emissions and removals.
- **Uncertainty Management:** Implements specific procedures for assessing the inherent uncertainty in quantification data, ensuring transparency and reliability of results obtained.

2.2 Continuous Monitoring of Environmental Performance in Agriculture

Provides the tools for continuous monitoring of the environmental performance of agricultural activities, enabling:

- **Assessment of the Impact of Practices:** Analyses the influence of different agricultural practices on GHG emissions and removals, making it possible to distinguish between practices that offer the greatest sustainability.
- **Tracking Progress:** Monitors the evolution of GHG emissions and removals over time, providing essential information for assessing the effectiveness of implemented mitigation strategies.
- **Identification of Areas for Improvement:** Supports the identification of specific areas within management processes that require improvement, enabling optimisation of the environmental performance of agricultural activities.

2.3 Clear and Consistent Reporting of Results

Defines principles and procedures for clear and consistent reporting of results, ensuring:

- **Transparency and Traceability:** Requires comprehensive documentation of the data collected and results achieved, ensuring maximum transparency and traceability of the quantification process.
- **Comparability:** Ensures data consistency over time, allowing direct comparison of environmental performance across different periods.
- **Compliance:** Promotes alignment with international standards for reporting GHG emissions and removals.

2.4 Support for the Transition to Sustainable Agricultural Systems

Provides practical support for the transition towards more sustainable agricultural systems, contributing to:

- **Identifying Best Practices:** Highlights the most effective agricultural practices for minimising emissions and maximising carbon sequestration.
- **Encouraging the Adoption of Sustainable Practices:** Incentivises the adoption of agricultural practices with a reduced environmental impact, offering the possibility of quantifying and assigning concrete value to the environmental benefits achieved.
- **Mitigating Climate Change:** Actively participates in global efforts to mitigate climate change, promoting the reduction of GHG emissions generated by the agricultural sector.

3. Applicability Conditions

To ensure the correct application of the TCR methodology, the following conditions must be met:

3.1 Scope of Application

This methodology is applicable to agricultural production systems that:

- **Include soil management:** The methodology is specifically designed for activities involving soil management, such as the cultivation of herbaceous and tree crops, the management of grasslands and pastures, and agroforestry.
- **Generate net GHG emissions and/or net GHG removals:** The methodology is applicable to agricultural systems that produce GHG emissions, remove GHGs from the atmosphere, or both. In particular:
 - **Net GHG reduction:** occurs when the implemented agricultural system produces lower emissions than the baseline, demonstrating that the mitigation intervention has contributed to a decrease in greenhouse gases (GHG). For example, the transition from conventional tillage to minimum tillage.
 - **Net GHG removal:** occurs when the agricultural system's capacity to remove greenhouse gases exceeds that of the baseline. In this case, the system absorbs a greater quantity of CO₂ from the atmosphere than in the reference scenario. An example is the integration of hedgerow borders with trees, which increases carbon sequestration.

- Are interested in quantifying and monitoring environmental impact: The methodology is aimed at organisations wishing to assess the environmental impact of their agricultural activities in terms of GHG emissions and removals, and who are interested in monitoring progress over time.

3.2 Data Availability

To correctly apply the methodology, sufficient and reliable data must be available on:

- Management practices: Detailed data on the management practices adopted, including types of crops grown, soil tillage techniques, use of fertilisers and pesticides, and crop residue management.
- Soil characteristics: Data on soil properties, including soil type, organic carbon content, texture and bulk density.
- Other relevant factors: Data on other factors that may influence GHG emissions and removals, such as climatic conditions, use of agricultural machinery and livestock management.
- The quality and reliability of the data collected will directly influence the accuracy and credibility of the quantification results.

3.3 Resources and Competencies

The application of this methodology requires:

- Adequate resources: Availability of qualified personnel, time and tools necessary for data collection, analysis and management.
- Specific competencies: Knowledge of agricultural practices, principles of GHG quantification and relevant standards.

3.4 Compliance with Standards

The methodology has been developed in compliance with international standards for the quantification, monitoring and reporting of GHG emissions and removals, including:

- ISO 14064-2: This standard provides the specifications and guidelines for quantifying, monitoring and reporting GHG emissions at project level.
- IPCC Guidelines: The IPCC guidelines for national greenhouse gas inventories are the primary reference for quantifying GHG emissions and removals.
- The adoption of additional standards relevant to the specific context, such as Verified Carbon Standards (VCS).

3.5 Limitations

The methodology has some inherent limitations, including:

- Availability of emission factors: The methodology relies on predefined emission factors for different agricultural activities. The availability of accurate, context-specific emission factors is fundamental to the accuracy of results.

- Complexity of agricultural systems: Agricultural systems are complex and influenced by numerous factors, which can make it difficult to accurately quantify GHG emissions and removals.
- Evolution of agricultural practices: Agricultural practices are constantly evolving, and this may require periodic updates to the methodology to ensure its relevance and accuracy.

4. Methodological Approach

This section describes in detail the phases, processes and procedures that make up the application of the TCR methodology.

4.1: System Definition and SSR Identification

- System delimitation: The system is defined as a farm or group of farms implementing sustainable agricultural practices to reduce emissions and increase GHG removals. The system boundaries include all agricultural activities carried out within the area identified during project registration (which may coincide with the entire farm or part of it; the land does not necessarily need to be contiguous), including crops, soil management and crop residue management. The time interval considered for analysis coincides with the duration of the monitoring period.
- SSR Identification: The identification of SSRs is based on the analysis of the agricultural practices implemented and the potential impacts of those practices on GHG fluxes. All SSRs affected by agricultural activities are considered, such as CO₂ emissions from agricultural machinery, N₂O emissions from nitrogen fertilisers, soil carbon sequestration and carbon absorption in plant biomass.
- SSR Classification: The identified SSRs are classified according to their relationship with the activities of the agricultural system. Controlled SSRs are those directly managed and influenced by the implemented agricultural practices. Related SSRs are those indirectly influenced by system activities, such as emissions associated with the production of fertilisers used in the system. Unaffected SSRs are those that are neither directly nor indirectly influenced by system activities.

4.2: Baseline Determination

- Baseline selection: The baseline represents the GHG emissions and/or removals that would occur if the conventional agricultural practices prevalent in the reference context continued to be implemented. This scenario, defined as 'business as usual' (BAU), is constructed by considering historical data on agricultural practices, sector trends and information from reliable sources.
- Baseline quantification: Baseline quantification is carried out using established methodologies and international standards, such as the IPCC Guidelines for National GHG Inventories and the Verra VM0042 standard. Calculations take into account activity-specific emission factors, soil characteristics and other relevant parameters, based on the data collected for the baseline.
- Uncertainty assessment: The uncertainty associated with the baseline estimate is assessed by considering the uncertainty of input data, the uncertainty of emission factors and the uncertainty of the quantification methodologies used. Standard techniques are applied for uncertainty propagation and determination of the confidence interval.

4.3: Quantification of Agricultural System Emissions and Removals

- **Data collection:** Data collection is based on a multi-level approach, combining data collected directly from farms (e.g., through questionnaires, interviews, farm records) with data from secondary sources (e.g., databases, scientific literature, statistics). The information collected concerns the agricultural practices implemented, soil characteristics (e.g., soil sampling and analysis), crop residue management, energy and fuel use, and other relevant parameters.
- **Application of quantification standards:** The quantification standards used to estimate agricultural system emissions and removals are those presented in Section 1.2, including LIFE C-Farms, Verra VM0042, CDM AR-AMS0007 and the IPCC Guidelines. The choice of specific methods depends on the SSRs considered and the agricultural practices implemented.
- **Uncertainty assessment:** The uncertainty associated with the estimation of agricultural system emissions and removals is assessed in a similar way to that described for the baseline, considering the uncertainty of input data, the uncertainty of emission factors and the uncertainty of the quantification methodologies used.

4.4: Calculation of Net GHG Reductions/Removals

- **Comparison with baseline:** The GHG emissions and/or removals estimated for the agricultural system (Phase 3) are compared with the baseline determined in Phase 2, using tonnes of CO₂ equivalent per year as the unit of measurement.
- **Calculation of net reductions/removals:** Net GHG reductions or removals are obtained by comparing the emissions and removals of the agricultural system with the baseline.
 - A net reduction when the emissions of the implemented agricultural system are lower than those of the baseline. This means that the mitigation intervention has led to a decrease in greenhouse gas (GHG) emissions. (For example, moving from conventional tillage to minimum tillage)
 - A net removal when the GHG removals by the implemented agricultural system are greater than those of the baseline. In this case, the agricultural system has absorbed more GHGs from the atmosphere than in the reference situation. (For example, implementing a system of borders with hedgerows and trees that sequester CO₂)
- **Uncertainty assessment:** The uncertainty associated with the calculation of net reductions/removals is determined by combining the uncertainty of the baseline estimate with the uncertainty of the agricultural system emissions/removals. Uncertainty propagation techniques are applied to obtain a confidence interval for net reductions/removals.

4.5: Monitoring and Reporting

- **Monitoring plan:** The monitoring plan defines monitoring activities, their frequency, data collection methodologies, indicators to be used and responsibilities. The plan takes into account the objectives of the quantification system, the characteristics of the agricultural system and the requirements of the reference standards.

- **Monitoring data collection:** Monitoring data collection is carried out through a combination of methods, including:
 - **In-situ monitoring:** Direct field measurements, e.g. soil sampling and analysis, biomass measurements, agricultural practice records.
 - **Remote sensing:** Use of satellite images to monitor changes in land cover, vegetation health and other relevant parameters.
 - **Databases and farm records:** Collection of data from farms, such as activity records, input consumption, production.
- **Analysis of monitoring data:** The monitoring data collected are analysed to assess GHG emission and removal trends, the effectiveness of the agricultural practices implemented and the potential need for adjustments to the management system. Statistical and spatial analysis techniques are used to interpret the data and identify trends and anomalies.
- **Reporting of results:** Results are reported through periodic reports documenting the data collected, the analyses carried out, the net GHG reductions/removals achieved and the associated level of uncertainty. Reports are prepared in accordance with reference standards and made public for verification activities.

4.6: Conversion of Net GHG Reductions/Removals into Carbon Credits

- **Verification and validation:** The data used to quantify net GHG reductions/removals are subjected to a verification and validation process by an independent third party.
- **Credit issuance:** Once quantification has been verified, carbon credits corresponding to the net GHG reductions/removals are issued according to the specifications of the reference standard. Each carbon credit represents one tonne of CO₂ equivalent removed or reduced from the atmosphere.

4.7: Notarisation of Carbon Credits

- **Registration:** Carbon credits are registered on a secure and transparent digital register, which tracks all information related to the credits, such as the project of origin, the quantity of credits issued, their status (e.g. available, retired) and transactions.
- **Blockchain notarisation:** Blockchain technology ensures the security and transparency of the carbon credit register, making information immutable and verifiable by all parties involved.

5. Quantification of GHG Emission Mitigations

5.1: Criteria and Procedures for Quantification

The CO₂ removals that can be generated by project activities are calculated as the difference between the project scenario (in which the virtuous practice is applied) and the standardised baseline. The difference (Δ) between these two scenarios corresponds to the quantity of CO₂ stored in the project pool and represents the additionality. The unit of measurement used is the tonne of carbon dioxide equivalent (tCO₂e).

The CO₂e removal activity must provide a net carbon removal benefit, which must be quantified using the following formula:

Net carbon removal benefit = $CR_{baseline} - CR_{total} - GHG_{increase}$ (eq.1) (Source: C-Farms)

where:

- $CR_{baseline}$ is the CO₂e removal relative to the baseline;
- CR_{total} is the total CO₂e removal from the carbon removal activity;
- $GHG_{increase}$ is the increase in direct and indirect greenhouse gas emissions, other than those from biogenic carbon reservoirs in the case of carbon-farming activities, that are due to the implementation of the carbon removal activity.

1. Carbon Removal Relative to the Standardised Baseline

The standardised baseline is identified with conventional management in cultivated land. The standard baseline varies according to the type of agricultural cultivation and must be representative of common practices for that specific crop.

For herbaceous crops, the standard baseline includes:

- Continuous cropping systems, with a single main crop per year.
- Monoculture, with a single plant species per field.
- Bare soil tillage, with deep ploughing that inverts soil layers.
- Use of a mouldboard plough.
- Removal of crop residues after harvest.
- Application of inorganic nitrogen fertilisers.

For tree crops, the standard baseline includes:

- Conventional management systems, with limited attention to biodiversity and soil health.
- Use of synthetic fertilisers and pesticides.
- Pruning and crop residue management with practices that do not maximise carbon sequestration (e.g. burning of residues).
- Little or no integration of agroforestry practices (e.g. hedgerows, rows, woods).

The standardised baseline is identified with conventional management in cultivated land, which includes continuous cropping systems, monoculture, bare soil tillage, mouldboard plough, removal of crop residues and application of inorganic nitrogen fertilisers.

Currently, data and methodologies are lacking to determine whether soils managed according to ordinary agricultural practices within project boundaries represent a net source or sink of CO₂. However, for a conservative standardised baseline, CO₂ emissions from organic carbon losses in cultivated soils can be considered equal to 0.

2. Carbon Removal Relative to Project Carbon-Farming Activities

CR_{total} , at the end of the monitoring period, is calculated on the basis of the measurement of carbon stocks at two points in time to assess changes in carbon storage due to the application of the carbon-farming practice. Carbon stocks include soil (SOC), living biomass

(LB) and harvested wood products (HWP) and are expressed in tonnes of CO₂e per hectare per year.

The change in carbon stocks in the project, occurring in the selected carbon reservoirs, in year t, is calculated as follows:

$$CR_{total} = \Delta CSOC + \Delta CLB + \Delta CHWP \quad (\text{eq.2}) \quad (\text{Source: C-Farms})$$

$$\Delta CSOC, LB, HWP = (C_{t1} - C_{t0}) / t1 - t0 \quad (\text{eq.3}) \quad (\text{Source: C-Farms})$$

$$\Delta CO_2 = -44/12 * \Delta C \quad (\text{eq.4}) \quad (\text{Source: C-Farms})$$

where:

- CR_{total} = Total change in carbon stocks of project carbon-farming activities, expressed in tonnes C yr⁻¹
- $\Delta CSOC$ = Total change in soil organic carbon stocks of project carbon-farming activities, expressed in tonnes C yr⁻¹
- ΔCLB = Total change in living biomass carbon stocks above and below ground of project carbon-farming activities, expressed in tonnes C yr⁻¹
- $\Delta CHWP$ = Total change in carbon stocks of harvested wood products from project carbon-farming activities, expressed in tonnes C yr⁻¹
- $\Delta CSOC, LB, HWP$ = Annual change in carbon stocks in the reservoir, in tonnes C yr⁻¹
- C_{t1} = Carbon stock in the reservoir at time t1, in tonnes C
- C_{t0} = Carbon stock in the reservoir at the beginning of the certification period (time t0), in tonnes C.
- ΔCO_2 = Annual CO₂ removals from net changes in soil carbon stock during the monitoring period, in t CO₂ yr⁻¹.

3. GHG Increase Relative to the Project Scenario

To calculate the increase in greenhouse gas emissions (GHG_{increase}), emissions from project carbon-farming activities must be compared with those generated in the baseline and included only when the project activity significantly increases such emissions relative to the baseline. The increase in GHG emissions can be generated by direct and indirect emission increases.

Therefore, the increase in GHG emissions (GHG_{increase}) is calculated through equation 5 (eq.5) and evaluates only differences > 0 derived from emissions between project carbon-farming activities and the baseline:

$$GHG_{increase} = GHG_{cf} - GHG_{bsl} \quad (\text{eq.5}) \quad (\text{Source: C-Farms})$$

$$GHG_{cf} = GHG_{direct} + GHG_{indirect} \quad (\text{eq.6}) \quad (\text{Source: C-Farms})$$

where:

- GHG_{increase} = increase in direct and indirect GHG emissions, other than those from biogenic carbon reservoirs in the case of carbon-farming activities [t CO₂e/yr]

- GHG_{bsl} = GHG emissions other than biogenic carbon reservoirs in the baseline [tCO₂e/yr], including soil emissions due to the application of fertilisers and the use of fossil fuels related to agricultural operations.
- GHG_{cf} = GHG emissions other than biogenic carbon reservoirs in the project scenario [tCO₂e/yr], including soil emissions due to the application of fertilisers and the use of fossil fuels related to agricultural operations.
- GHG_{direct} = direct GHG emissions, other than biogenic carbon reservoirs, due to carbon-farming activities within project boundaries [tCO₂e/yr]
- GHG_{indirect} = direct GHG emissions, including biogenic carbon reservoirs, due to carbon-farming activities outside project boundaries [tCO₂e/yr]

$GHG_{cf,bsl} = GHG(INF) + GHG(FUEL) + GHG(OA) + GHG(CC)$ (eq.7) (Source: C-Farms)

$GHG(INF) = X(INF) \times EF(INF) / 1000$ (eq.8) (Source: C-Farms)

$GHG(FUEL) = X(FUEL) \times EF(FUEL) / 1000$ (eq.9) (Source: C-Farms)

$GHG(OA) = X(OA) \times EF(OA) / 1000$ (eq.10) (Source: C-Farms)

$GHG(CC) = X(CC) \times EF(CC) / 1000$ (eq.11) (Source: C-Farms)

where:

- GHG_{cf,bsl} = total emissions of the baseline scenario or project scenario, expressed in tCO₂/ha/yr
- GHG(INF) = direct and indirect soil emissions due to the application of inorganic nitrogen fertilisers, expressed in t CO₂/ha/yr
- GHG(FUEL) = direct emissions from the use of fossil fuels for machinery operations, expressed in t CO₂/ha/yr
- GHG(OA) = direct and indirect soil emissions due to the application of organic nitrogen fertilisers, expressed in t CO₂/ha/yr
- GHG(CC) = direct and indirect soil emissions due to the cultivation of nitrogen-fixing cover crops with return of biomass to soil, expressed in t CO₂/ha/yr
- X = quantity of nitrogen applied to soil, expressed in kg N/ha/yr
- EF_{INF,OA,CC} = emission factor for the specific fertiliser, expressed in kg CO₂eq/kg N (Table 7 from the C-Farms reference document)
- EF_{FUEL} = emission factor for diesel consumption, expressed in kg CO₂eq/l (Table 8 from the C-Farms reference document)

4. Leakage

Leakage is defined as the net change in GHG emissions outside project boundaries. The TCR methodology promotes the implementation and intensification of sustainable agricultural practices, anticipating that the implementation of such practices will increase agricultural production in the regions, minimising emissions outside project boundaries.

5. Reserve (Permanence Buffer)

To ensure the permanence of generated credits over time, a buffer system has been established, in which a percentage of the carbon absorption units generated is set aside to

guarantee the permanence of generated credits over time. This percentage has been set at 10% of carbon removal units to cover any losses (Buffer).

6. Baseline

6.1 Description

The baseline represents the situation that would occur in the absence of mitigation interventions to reduce emissions and increase carbon sequestration in the agricultural sector. In this scenario, conventional agricultural practices continue to prevail, characterised by:

- Widespread use of synthetic inputs: Strong dependence on synthetic fertilisers and pesticides.
- Limited organic matter recycling: Low attention to reuse of crop residues and incorporation of organic matter into soil.
- Practices that promote soil degradation: Intensive soil tillage techniques and low attention to soil health, resulting in reduced carbon sequestration capacity.
- In this context, GHG emissions from agricultural activities remain at high levels or could increase due to continued dependence on synthetic inputs and progressive degradation of soil health.

6.2 Justification

The choice of the baseline as 'business as usual' (BAU) is justified by the widespread use of conventional agricultural practices in the reference agricultural sector. This situation is supported by historical data analysis, sector trends and information from authoritative sources. The BAU scenario provides a realistic reference point for quantifying the environmental benefits of mitigation interventions, enabling assessment of the positive impact of sustainable practices relative to the current situation.

6.3 Assumptions

The baseline is based on the following assumptions:

- Persistence of conventional practices: In the absence of targeted interventions, conventional agricultural practices will continue to be widely adopted.
- Absence of significant incentives: No economic or regulatory incentives are foreseen that are sufficiently strong to induce significant changes in agricultural practices.
- Limited spontaneous adoption of sustainable practices: It is assumed that the adoption of sustainable agricultural practices by farmers, without external incentives, will remain limited.

6.4 Lack of Incentives for Change

In the reference context, stable and long-term economic and regulatory incentives that encourage farmers to adopt more sustainable agricultural practices are lacking. This situation limits the spread of practices for reducing emissions and increasing carbon sequestration, making targeted programmes and policies necessary to promote the transition to more sustainable agriculture.

6.5 Additionality

Considering the persistence of conventional practices and the lack of incentives for change, it is assumed that mitigation interventions, based on the implementation of sustainable agricultural practices, will generate additional net GHG reductions/removals relative to the baseline. This means that mitigation actions will lead to a quantifiable improvement in environmental performance relative to the BAU situation.

7. GHG Sources, Sinks and Reservoirs

This section defines the greenhouse gas (GHG) sources, sinks and reservoirs relevant to the methodology for quantifying, monitoring and reporting emissions and removals in the agricultural sector. The choice of elements to include is based on the analysis of agricultural practices and their potential impacts on GHG fluxes, taking into account international standards and the quantification methodologies used.

7.1 Identification of Carbon Reservoirs

The following table identifies the carbon reservoirs considered by the methodology, specifying whether they are included or excluded from the application boundary, both in the baseline and in the scenario with implementation of the methodology.

Table 7a: Identification of Carbon Reservoirs

Carbon Reservoirs	Justification / Explanation	Included/Excluded
Living biomass (above and below ground)	Living biomass must be included where activities involve an increase in this reservoir (plantations, agroforestry). Removal of woody biomass due to activities is excluded from applicability conditions.	Included
Waste/Debris	Must be included where activities could significantly reduce the reservoir. Excluded because it is not subject to significant variations or potential variations are transitory in nature on agricultural land.	Excluded
Dead Wood	Included because activities could significantly reduce the reservoir. Pruning plays a central role in the context of woody perennial species and is a key management activity that produces on average 3 tonnes of material per hectare each year.	Included
Soil Organic Carbon (SOC)	Included – Main carbon reservoir affected by activities that is expected to increase with implementation of the methodology.	Included
Harvested Wood Products	Included – Although no woody products are expected to be harvested, this carbon reservoir will be monitored to ensure long-term carbon storage.	Included

7.2 Identification of Greenhouse Gases

The following table identifies the greenhouse gases (GHGs) associated with the carbon reservoirs considered by the methodology, specifying whether they are included or excluded from the application boundary, both in the baseline and in the scenario with implementation of the methodology.

Table 7b: Identification of GHGs of Carbon Reservoirs

Carbon Reservoirs	GHG	Justification / Explanation	Included/Excluded
Soil Organic Carbon (SOC)	CO ₂	Included – Expected to be the most significant reservoir affected by activities. The methodology does not require accounting for CO ₂ emissions from fossil fuels if activities will not significantly increase emissions relative to baseline.	Included
Fossil Fuels	CO ₂	Excluded – Adoption of practices such as no-till is expected to reduce fossil fuel emissions from tractors. Therefore, excluding this GHG SSR is considered conservative.	Excluded
Soil Methanogenesis	CH ₄	Excluded – Soil methanogenesis is more common in flooded cropping systems such as rice and is therefore negligible in the cropping systems and climate zones considered in this methodology.	Excluded
Enteric Fermentation	CH ₄	Excluded – Although livestock management is not a targeted activity, activities may involve operators who incorporate livestock into their system in the baseline and with the methodology. As this practice is not widespread, it is temporarily excluded.	Excluded
Manure Deposition	N ₂ O	Excluded – Similarly to enteric fermentation, this practice is not yet widespread and is therefore excluded.	Excluded
Use of Nitrogen Fertilisers	N ₂ O	Included – Reduction of synthetic nitrogen fertilisers is part of the activities considered.	Included
Use of Nitrogen-Fixing Species	N ₂ O	Included – Nitrogen-fixing species are considered by the methodology.	Included
Biomass Combustion	CO ₂	Excluded – No biomass combustion is permitted as an applicability condition.	Excluded
	CH ₄	Excluded – No biomass combustion is permitted as an applicability condition.	Excluded
	N ₂ O	Excluded – No biomass combustion is permitted as an applicability condition.	Excluded

8. Example of Determination of Sampling and Monitoring Areas

Carbon measurement is a vital part of carbon credit projects because emission reductions and removals from plantation sites are estimated by measuring changes in the quantity of carbon, and credits are also calculated using sequestered carbon. Measurement can be defined as the continuous measurement and collection of data on anthropogenic GHG emissions related to forests from sources and on removals from sinks. The measurement system must be transparent, consistent and accurate, and uncertainty must be minimised.

8.1 Carbon Reserves

The IPCC (2006) defines five carbon reserves:

1. Above-Ground Biomass
 2. Below-Ground Biomass
 3. Litter
 4. Dead Wood
 5. Soil Carbon
- Above-Ground Biomass: All biomass of living vegetation, both woody and herbaceous, above ground, including stems, stumps, branches, bark, seeds and foliage.
 - Below-Ground Biomass: All biomass of living roots. Fine roots of diameter less than (suggested) 2mm are often excluded because they often cannot be empirically distinguished from soil organic matter or litter.
 - Litter: Includes all non-living biomass with a size greater than the limit for soil organic matter (suggested 2mm) and less than the minimum diameter chosen for dead wood (e.g. 10cm), lying dead, in various states of decomposition above or within mineral or organic soil. This includes the litter layer as usually defined in soil types. Living fine roots above mineral or organic soil (of diameter less than the minimum limit chosen for subsurface biomass) are included in litter where they cannot be empirically distinguished from it.
 - Dead Wood: Includes all non-living woody biomass not contained in litter, whether standing, lying on the ground or in the soil. Dead wood includes wood lying on the surface, dead roots and stumps, with diameter greater than or equal to 10cm (or the diameter specified by the country).
 - Soil Organic Carbon: Organic carbon in mineral and organic soils (including peat) at a specified depth chosen by the country and applied consistently throughout the time series. Living fine roots of diameter less than 2mm (or other value chosen by the country as the diameter limit for below-ground biomass) are included with soil organic matter where they cannot be empirically distinguished from it. Soil samples will be collected from a depth of 30 cm.

The IPCC (2003b) also provides three tiers for estimating emissions, with increasing levels of data requirements and analytical complexity and therefore greater precision:

- Tier 1 uses predefined emission factors (indirect estimation of emissions based on reduction of canopy cover) for forest activities ('activity data') collected at national or global level.
- Tier 2 applies emission factors and activity data from country-specific data.
- Tier 3 uses methods, models and inventory measurement systems that are repeated over time.

8.2 Carbon Estimates in Different Stocks

I. Above-Ground Biomass

Above-ground biomass includes the biomass of living trees and non-tree biomass comprising grasses and shrubs.

a) Living Tree Biomass:

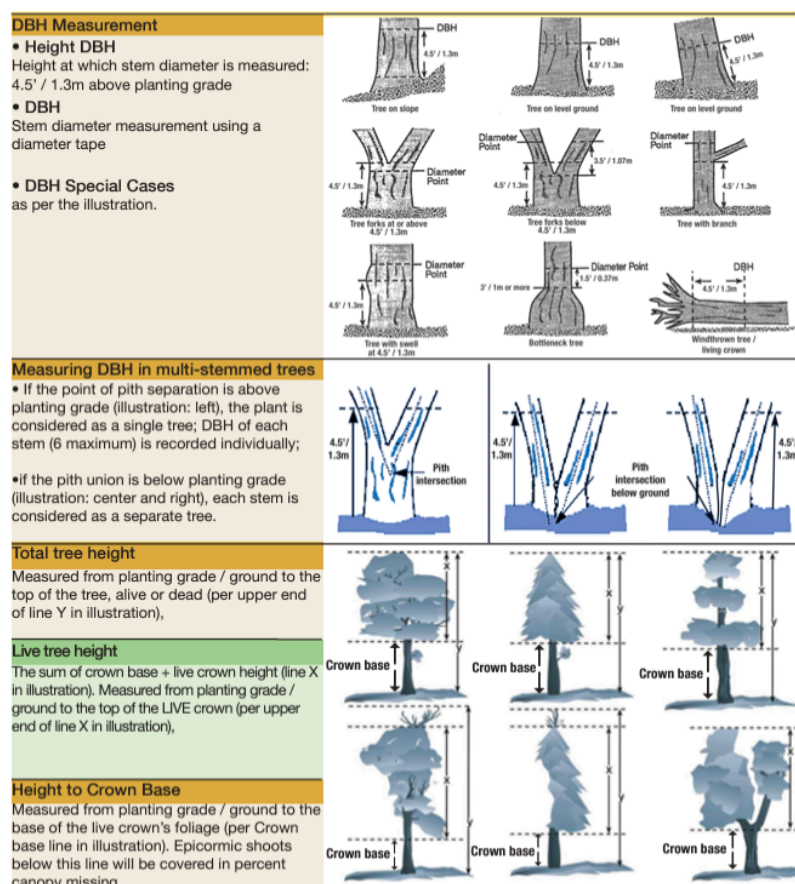
Tree biomass is generally estimated using developed volumetric equations. Volumetric equations relate biomass to tree height and/or Diameter at Breast Height (DBH) measured at 1.37 m from the ground. Measurement of tree parameters can be carried out following these steps:

Field Work

6. Identify the tree species.
7. Measure the DBH and height of all trees in sample plots according to the procedure described below:
 - DBH Measurement: Position the tape around the trunk at the 1.37 m mark. Measure the DBH at 1.37 m from the ground. DBH measurement depends on terrain slope, tree lean angle and trunk shape (see figure below).
 - Slope: Measure the diameter on the uphill side of the tree.
 - Leaning tree: Always measure the height of one measurement (1.37 m) parallel to the tree, not perpendicular to the ground. Therefore, if the tree is leaning, measure under the lean, parallel to the tree's angle.
 - Multi-stemmed tree: If the tree has multiple stems with forks below the measurement point, measure the diameter on each stem and mark the stems. Record each stem as if it were a different tree in the data sheet, with a note that the stems constitute a single tree.
 - Height Measurement: Tree height can be measured using various instruments such as the clinometer and laser rangefinder.
 - Stand far enough from the tree so that the top of the tree is less than 90 degrees relative to the line of sight.
 - Always stand on the uphill side of the tree. Standing on the downhill side should only be done when there is no other option.
 - Measure the height of the dominant crown of the tree.
8. Place a chalk mark on the tree to indicate that the tree has been measured.
9. All trees should be tagged with an identification tag.
10. Record the species name with the local name and the corresponding DBH and height in the format.

(Source: i-Tree

https://www.itreetools.org/documents/662/Cheat_Sheet_English_for_distribution_03.01.2021_OFS.pdf)



(Source: i-Tree – DBH Measurement Guide)

Post-Field Work

11. Tree Volume Estimation: Estimate the volume of each tree in the sample plot using volumetric or allometric equations.
12. Obtain Wood Density: Obtain wood density for all tree species encountered in the sample plot from the literature.
13. Calculation of Dry Weight of each Tree: Multiply the volume of each tree by the corresponding wood density to obtain the dry weight of each tree.
14. Multiplication by Biomass Expansion Factor: Multiply the obtained dry weight of each tree by the Biomass Expansion Factor (BEF).
15. Sum of Weights of all Trees of each Species in the selected Sample Plots.
16. Sum of Weights of all Trees of all Tree Species for all Sample Plots.
17. Extrapolation of the Weight of each Species from the total sampled area (sum of all plots) to value per hectare (tonnes of biomass per hectare per species).
18. Sum of biomass of each species to obtain the total biomass of all trees in tonnes per hectare.
19. Carbon is 47% of dry biomass.

b) Grass Biomass:

Grass sampling is carried out by establishing 1×1 metre plots. The following steps can be followed for estimating grass biomass:

20. Recording the species name of each grass in the format.

21. Collection of all grasses in the plot.
22. Fresh weight of each species must be measured using a portable scale and recorded in the format.
23. The sample must be adequately packaged and taken to the laboratory for further estimates.
24. Estimation of the dry weight of the sample by drying it in an oven.
25. Extrapolation of the sample plot value of dry grasses to stock per hectare.

c) Shrub Biomass:

26. 3×3 metre plots will be laid out for estimating shrub biomass. All shrubs in the plot will be collected and weighed. The sample should be taken to the laboratory for further biomass estimates. The sample must be dried to constant weight in an oven and values extrapolated to stock per hectare.

II. Below-Ground Biomass:

The Root-Shoot Ratio of 0.26 indicates the relationship between root biomass and above-ground biomass of a tree. This ratio is used to estimate Below-Ground Biomass (BGB, Cairns et al. 1997). For example, if the above-ground biomass of a tree is known, below-ground biomass can be estimated by multiplying the above-ground biomass by the Root-Shoot ratio of 0.26. This approach provides an estimate of the distribution of tree biomass between above-ground parts and roots, based on empirical data and observed relationships.

III. Litter:

The steps for measuring litter biomass are as follows:

27. Collect litter present in the sample plot.
28. Record the fresh weight of total litter collected.
29. Take a 200-gram sample of litter.
30. The sample must be appropriately marked and packaged. Take the sample to the laboratory for further estimates.
31. Dry the litter sample in a laboratory oven until constant weight and weigh the dried sample.
32. Extrapolate the sample plot value of dried litter to stock per hectare.

IV. Dead Wood:

Dead wood includes all non-living woody biomass not contained in litter, whether standing, lying on the ground or in the soil (Ravindranath and Ostwald, 2008). Dead wood biomass is measured in the same way as living trees.

V. Soil Organic Carbon;

Precautions: During sampling, it is important to keep in mind the following:

- Position sampling site away from roads, houses, construction sites, etc.
- The sample should be taken away from tree trunks or between trees.
- Avoid eroded areas and places where large plant material is decaying.

- Always dig a new rectangular pit and, in grasslands, first remove the top layer and dig the profile.

Sample Collection: At each sampling point, a soil sample is collected from a depth of 0–30 cm.

Soil sample collection for carbon estimation: Remove litter from an area of 1.0 m × 1.0 m at the sampling point, and dig a pit 30 cm wide, 30 cm deep and 50 cm long. Soil from the three sides of the pit, from the surface to 30 cm depth, is scraped and collected. This soil is thoroughly mixed and approximately 200 grams are taken. The soil sample is stored in a polyethylene bag, hermetically sealed and correctly labelled.

Soil sample collection for bulk density estimation: Insert the bulk density cylinder (of known volume) between 0–15 cm depth with the help of a hammer, until the top of the cylinder. Carefully remove the cylinder so that the soil inside does not fall out. Collect all the soil in the polyethylene bag and take to the laboratory. Repeat this operation at soil depth between 15 and 30 cm. Store these soil samples in an oven at 65°C until completely dried. Measure the dry weight of the soil and calculate soil bulk density.

Soil bulk density is calculated using the following formula:

Soil bulk density = Dry weight of soil (g) / Volume of cylinder (cm³)

This calculation gives the measure of soil bulk density based on the dry weight of soil and the volume of the cylinder used during measurement. Finally, the average of the two densities indicated above will be calculated to obtain the average soil weight for that particular site.

Estimation of percentage of coarse fragments in soil: The percentage of coarse fragments (>2 mm size) in soils is estimated to correctly calculate soil weight. After taking the weight of the dried sample for bulk density, this sample is placed in a 2 mm sieve and water is run over it. Soil particles smaller than 2 mm will be carried away with the water. Subsequently, the fraction is removed from the sieve, dried and weighed. The percentage of coarse fragments is then calculated.

Sample preparation for soil organic carbon estimation: Open the polyethylene bag and distribute the samples on a sheet of brown paper in the laboratory. Allow the sample to air-dry, avoiding direct sun or oven drying. After drying, grind the sample and sieve through a 2 mm sieve. This sieved sample is used for soil organic carbon estimation.

Sample analysis for soil organic carbon: The percentage of soil organic carbon is estimated according to the standard Walkley & Black (1934) method.

Soil carbon calculations: Soil stoniness and land use are recorded. Soil samples are analysed for required parameters such as bulk density, organic carbon and texture. The estimate of soil organic carbon, Q_i (Mg m⁻²), in a soil layer or sampling level i with a depth of E_i (m), depends on carbon content C_i (g C g⁻¹), bulk density D_i (Mg m⁻³) and the volume fraction of coarse elements G_i , as indicated by the formula of Batjes (1996).

Total density of stored carbon: The density of stored carbon is calculated by summing the densities of stored carbon in the individual carbon reservoirs of that layer using the formula:

TC = Above-ground Carbon (Living Trees + Grasses + Shrubs) + Below-ground Carbon + Litter Carbon + Dead Wood Carbon + Soil Organic Carbon

8.3 Delimitation of Project Spatial Boundaries

The spatial boundary of the project is the geographically delimited area dedicated to project activity. The Keyhole Markup Language (KML) file format must be provided with the Project Design Document. Each file must be accompanied by a clear description, and the metadata must contain all the necessary projection reference data.

8.4 Role of Remote Sensing and Geographic Information Systems (GIS)

Remote sensing provides a method for determining biomass through a combination of remote sensing data and field measurements to provide an accurate and cost-effective estimate of biomass over a large project area. Project developers must be able to obtain remotely acquired satellite data covering the area of interest. Accurate past land use and land cover (LULC) data in the project area must be available for at least three points in time, with at least one remote sensing image from 0–3 years before the project start date, at least one image from 4–9 years before the project start date, and at least one image from 10–15 years before the project start date. Images older than 15 years may not be used for the historical reference period.

6.4.1 Pre-processing: Raw remote sensing data may exhibit various distortions. Satellite data must be pre-processed to correct radiometric and geometric distortions. Radiometric correction reduces the impact of atmospheric conditions. Geometric correction of the image is the process of digital manipulation of image data to avoid geometric distortions so that the image projection precisely matches a specific surface or projection shape.

6.4.2 Ground Truthing: Field surveys can be used to collect and record information on land use, and to be used as independent ground-truth data for remote sensing classification. Field survey data on land covers, planting sites, tree species, agroforestry practices, year of planting collected to verify remote sensing data are called Ground Truthing. Field verifications can be collected via random sampling.

8.5 Sampling Design

The sampling design aims to identify sample points in each selected stratum. Sampling includes the decision on the number, size and shape of plots. It is crucial for measuring and monitoring changes in carbon stocks. A carbon inventory based on appropriate sampling can provide reliable estimates at limited cost and human effort. Sampling allows conclusions to be drawn about an entire population by observing only a part of it. Sampling theory provides the means to scale information from sample plots to the entire project area or even at regional and national level (IPCC 2003). Two statistical concepts are involved in sampling:

Accuracy: is a measure of how close the sample measurements are to actual values.

Precision: is a measure of how well a value is defined. Accuracy and precision reflect how well measurements estimate the true value of tree variables such as diameter, height and area covered by tree vegetation.

The determination of the sample size (number of sample plots) required for strata depends on: i) The required precision of at least 15% at a 95% confidence level. ii) The expected variance in the specific stratum.

8.6 Permanent Sample Plot

Permanent sample plots are generally considered statistically more efficient in estimating changes in carbon stocks compared to temporary plots because there is typically a high covariance between observations made at successive sampling events in temporary plots. Permanent sample plots will be established for the assessment and monitoring of carbon stocks in the project area.

The CDM A/R methodological tool "CDM A/R – Calculation of the number of sample plots for measurements within A/R CDM project activities" can be used to determine the required number of biomass sample plots.

8.7 Calculation of the Number of Permanent Sample Plots

33. The number of sample plots required for estimating biomass stocks in a carbon pool depends on the targeted precision and the variability of the estimated biomass.
34. The project area is stratified based on the variability of estimated biomass, and the approximate area of each stratum is determined. If estimated biomass is the sum of biomass stocks in two or more pools, then stratification is performed based on the variability of the dominant pool biomass (i.e., the pool containing the greatest amount of biomass).
35. For the purpose of this tool, the variability of biomass stocks is expressed as the standard deviation of biomass stocks in the stratum. The approximate value of the standard deviation of biomass stocks in each stratum at the time of estimation is known from existing data applicable to the project area or existing data relating to a similar area, or is estimated based on a preliminary sample or expert judgement.
36. The number of sample plots required for estimating biomass stocks within the project boundary is calculated iteratively as:

$$n = \frac{N * t_{VAL}^2 * \left(\sum_i w_i * s_i \right)^2}{N * E^2 + t_{VAL}^2 * \sum_i w_i * s_i^2}$$

(Source: UNFCCC – CDM – A/R Methodological Tool, Calculation of the number of sample plots for measurements within A/R CDM project activities – Section I/II (Version 02.1.0))

Where:

n = Number of sample plots required for estimating biomass stocks within the project boundary; dimensionless

N = Total number of possible sample plots within the project boundary (i.e., sampling space or population); dimensionless

t_{VAL} = Student's t-value (two-sided), at infinite degrees of freedom, for the required confidence level; dimensionless

w_i = Relative area weight of stratum i (i.e., the area of stratum i divided by the project area); dimensionless

s = Estimated standard deviation of biomass stocks in stratum i; t d.m. (or t d.m. ha⁻¹)

E = Acceptable margin of error (i.e., half the confidence interval) in estimating biomass stocks within the project boundary; t d.m. (or t d.m. ha⁻¹), i.e., in the units used for s_i.

i = 1, 2, 3, ... Biomass stock estimation strata within the project boundary.

8.8 Assignment of Number of Sample Units Among Strata

The optimal allocation method is used to distribute the number of sample units among the different strata. The number of sample units assigned to each stratum is calculated as:

$$n_i = n * \frac{w_i * s_i}{\sum_i w_i * s_i}$$

(Source: UNFCCC – CDM – A/R Methodological Tool, Calculation of the number of sample plots for measurements within A/R CDM project activities – Section I/II (Version 02.1.0))

Where:

n_i = Number of sample units allocated to stratum i; dimensionless

n = Number of sample units required for estimating biomass stocks within the project boundary; dimensionless

w_i = Relative area weight of stratum i (i.e., the area of stratum i divided by the project area); dimensionless

s_i = Estimated standard deviation of biomass stocks in stratum i; t d.m. (or t d.m. ha⁻¹)

i = 1, 2, 3, ... biomass stock estimation strata within the project boundary

8.9 Monitoring Plan

A monitoring plan will be established to obtain, record, compile and analyse the important data and information for quantifying and reporting the GHG emissions and/or removals relevant to the project (including the risk of losses) and the baseline.

8.10 Reporting

'Reporting' means providing information on estimates of emissions and removals, estimation methods, procedures and systems, including current conditions and future projections in accordance with the reporting guidelines, formats and procedures prescribed to designated organisations. Reporting enables comparison of emission reductions and removals and ensures transparency (FFPRI, 2010). Five fundamental principles for reporting GHG emissions and removals are transparency, accuracy, comparability, completeness and consistency. These principles should be followed before presenting reports to interested organisations. Quantitative information on GHG emission and removal measures, natural environment measures including flora and fauna, social and economic status of communities and the estimation process is required for reliable reporting. The credibility of reporting includes compliance with safeguards and ensures environmental and livelihood benefits, as well as the integrity of carbon measurements.

8.11 Verification

Verification in the context of MRV of GHG emissions and removals is a procedure for assessing and ensuring, from an independent perspective, the reliability of reporting of estimates and the adequacy of methods, procedures and documentation. Verification in carbon finance projects is part of the ex-post evaluation process that confirms that the emission reductions obtained from the project and the carbon credits are reliable. Verification will be carried out by a third party to confirm that the project has been conducted as prescribed in the project plan. The verification report describes the verification process, any findings made during the verification and the related solutions, as well as the conclusions reached by the validation/verification body.

9. Permanence Risk Assessment

In order to validate the applicability of the TCR methodology, the Verra "AFOLU Non-Permanence Risk Tool" Version 4.2 (October 2023 version) was applied to a reference project in which reforestation activities would be applied over a period of 20 years. This tool assesses the internal risk of a project, external risk, natural risk and mitigation measures that help reduce risk.

The following report demonstrates step by step the path taken to obtain a consistent risk assessment, considering the three risk categories present in the guidelines, as follows.

9.1 Internal Risks

	Project Management	Score
Q1	Does the project have an adaptive management plan that includes a monitoring plan? Answer: Yes	
Q2	"One or more of the following management risks apply to the project?" Answer: f, g	
a)	Planted species (where applicable) associated with more than 25 percent of stocks on which GHG credits have previously been issued are neither native nor demonstrated to be adapted to the same or similar agro-ecological zone(s) of the project area.	0
b)	Constant control is required to prevent intrusion by external actors in order to protect more than 50% of stocks on which GHG credits have previously been issued.	0
c)	The management team does not include individuals with significant experience in all the competencies necessary to successfully carry out the project activities (i.e., no area of required expertise is covered by an individual with at least five years of experience in that area).	0
d)	The management team does not maintain a presence in the country or is located more than one day's travel from the project site, considering all parcels or polygons in the project area.	0
e)	The management team has not previously submitted a loss report within two years of detecting a loss event.	0
f)	ALM Projects: Some or all of the farmers participating in the project have not received training on the improved agricultural land	2

	Project Management	Score
	management practices implemented as part of the project or on the monitoring and reporting procedures implemented during the crediting period.	
g)	ALM Projects: Some or all of the farmers participating in the project are not aware of the possibility that yields may temporarily decrease due to the transition to improved agricultural practices.	2
Q3	One or more of the following mitigations apply to the project? Answer: h, i	
h)	Mitigation: The management team includes individuals with significant experience (i.e., more than five years) in the design and implementation of AFOLU projects, carbon accounting and reporting (e.g., individuals who have successfully managed projects through validation, verification and issuance of GHG credits) under the VCS Programme or other approved GHG programmes.	-2
i)	Mitigation (ALM Projects): A comprehensive training plan is in place for all project-participating farmers, covering implementation of the intended agricultural land management practices, monitoring and reporting obligations and associated potential costs.	-2
	Total Project Management (PM) = [as applicable, (a + b + c + d + e + f + g + h + i)]The Total must not be less than zero	0

	Financial Viability	Score
Q1	How long is the project payback period (i.e., how many years will it take to break even)? Answer: e	
a)	The payback period is more than 20 years according to the current risk assessment.	Risk Failed
b)	The payback period is more than 10 years according to the current risk assessment, but less than or equal to 20 years.	0
c)	The payback period is more than seven years and up to ten years inclusive, according to the current risk assessment.	0
d)	The payback period is more than four years and up to seven years inclusive, according to the current risk assessment.	0
e)	The payback period is four years or less, according to the current risk assessment.	0
Q2	What percentage of funding has the project secured to cover the total money needed before the project breaks even? Answer: f	
f)	The project has secured less than 15 percent of the funding needed to cover the total money required before the project breaks even.	3
g)	The project has secured from 15 percent to less than 40 percent of the funding needed to cover the total money required before the project breaks even.	0
h)	The project has secured from 40 percent to less than 80 percent of the funding needed to cover the total money required before the project breaks even.	0

	Financial Viability	Score
i)	The project has secured 80 percent or more of the funding needed to cover the total money before the project breaks even.	0
Q3	Does the following mitigation measure apply to the project? Yes	
j)	Mitigation: The project has available, callable and secured financial resources equal to at least 80 percent of the total money needed before the project breaks even.	-2
	Total Financial Viability (FV) = [as applicable, ((b, c, d or e) + (f, g, h or i) + j)]The total must not be less than zero and the subtotal for question two with mitigation "j" must not be less than zero (note: the mitigation applies only to question 2).	1

	Opportunity Cost	Score
Q1	Are baseline activities driven by subsistence? If No, proceed with question 2. If Yes, proceed with question 3. Answer: No	
Q2	What is the NPV of the most profitable alternative land use activity compared to the NPV of the project activity? An answer is only required if the answer to question 1 is No. Answer: c	
a)	The net present value (NPV) of the most profitable alternative land use activity is expected to be at least 100 percent higher than that of the project activities.	0
b)	The net present value (NPV) of the most profitable alternative land use activity is expected to be more than 50 percent and up to 100 percent higher than that of the project activities.	0
c)	The net present value (NPV) of the most profitable alternative land use activity is expected to be more than 20 percent and up to 50 percent higher than that of the project activities.	4
d)	The net present value (NPV) of the most profitable alternative land use activity is expected to be 20 percent or less than that of the project activities.	0
e)	The net present value (NPV) of the project activities is expected to be more profitable by 20 percent and up to 50 percent than that of the most profitable alternative land use activity.	0
f)	The net present value (NPV) of the project activities is expected to be more profitable by more than 50 percent than that of the most profitable alternative land use activity.	0
Q3	Does the project have net positive impacts on the community? An answer is required only if the answer to question 1 is Yes.	
g)	The net positive impacts on the community of the project activities are not demonstrated.	0
h)	The net positive impacts on the community of the project activities are demonstrated.	0
Q4)	One or more of the following mitigations apply to the project? Answer: i	
i)	Mitigation: The project is protected by a legally binding agreement (see Section 2.2.5) to continue the management practices that protect	-2

	Opportunity Cost	Score
	the accredited carbon stocks for the entire duration of the project crediting period.	
j)	Mitigation: The project is protected by a legally binding agreement (see Section 2.2.5) to continue the management practices that protect the accredited carbon stocks for at least 100 years.	0
k)	Mitigation: Where there is the potential for a revenue loss relative to the most profitable alternative land use activity, the project is non-profit or has additional financial support (e.g., through grants, government funding, payments for ecosystem services or SD VISTA activities) to overcome the expected revenue loss.	0
	Total Opportunity Cost (OC) = [as applicable, ((a, b, c, d, e, f, g or h) + (i or j) + k)]The total must not be less than zero.	2

	Project Longevity	Score
Q1	Did the project request registration on 1 January 2024 or later? Answer: Yes	
Q2	Does the project have a legally binding agreement covering at least a period of 100 years from the project start date? If Yes, the project is assigned a zero for this risk category. If No, proceed with Q3. Answer: No	
Q3	What is the duration of the project in years? If you answered Yes to question 1 and the project duration is less than 40 years, the project fails the risk assessment. If you answered No to question 1 and the project duration is less than 30 years, the project fails the risk assessment. Note that projects with a duration of less than 40 years will not be eligible for the Core Carbon Principles label.	40
Q4	Does the project have a management, financing and monitoring plan for the entire duration of the project? If No, the project fails the risk assessment. If Yes, proceed with Q5. Answer: Yes	
Q5	Is the project an afforestation, reforestation and carbon dioxide removal (ARR) or sustainable forest management (IFM) with harvesting project? If No, proceed with Q7. If Yes, proceed with Q6. Answer: Yes	
Q6	Can the project demonstrate a commitment to continue the management practice, replant or allow regrowth? If No, the project fails the risk assessment. If Yes, proceed with Q7. Answer: Yes	
Q7	Does the project have a legal agreement or requirement to continue the management practice? If No, proceed with a. If Yes, proceed with b. Answer: Yes	
a)	Without a legal agreement or requirement to continue the management practice.	0
b)	With a legal agreement or requirement to continue the management practice.	15
Q8	Is the project a grouped project in which the durations of contracts with the individual project activity instances are less than the project duration? Answer: No	

	Project Longevity	Score
	Total Project Longevity (PL) = [as applicable, (a or b)]The Total must not be less than zero	15

Total Internal Risk
Total Internal Risk = (PM + FV + OC + PL)
The Total must not be less than zero

9.2 External Risks

	Land Tenure (Ownership) and Access to or Impacts on Resources	Score
Q1	Has due process been followed to discover any disputes over ownership and access or resource use rights? If the answer is No, the project fails the risk assessment. If the answer is Yes, proceed with Q2. Answer: Yes	
Q2	Has the project entered into a legally binding agreement (such as a contract) that guarantees the legal right to control and manage the project activities over the entire project area with all entities that have verified ownership claims or verified rights of access or resource use (such as holders of customary rights)? If the answer is No, the project fails the risk assessment. If the answer is Yes, proceed with Q3. Answer: Yes	
Q3	Do the same or different entities hold land ownership and resource access or use rights? Answer: Same	
a)	Land ownership and resource access or use rights are held by the same entity (or the same entities).	0
b)	Land ownership and resource access or use rights are held by different entities (e.g., the government owns the land and the project proponent holds a lease or concession).	0
Q4	Is the project located in a country/jurisdiction with a history of national, sub-national or local government intervention in land or resource management? Answer: e	
c)	The government has previously expropriated significant areas of land (i.e., 10% or more) in the project area in the last 20 years.	0
d)	The government has previously modified land property rights in the project jurisdiction (e.g., cancelled or blocked property titles, expropriated land or issued overlapping property titles) in the last 20 years.	0
e)	No cases of government intervention in the last 20 years or specific cases of expropriation and government intervention in land rights in the project area have been definitively resolved against the government in a competent court.	0
Q5	What percentage of the project area is affected by disputes regarding land possession or ownership? Answer: h	

	Land Tenure (Ownership) and Access to or Impacts on Resources	Score
f)	Disputes exist in more than 5 percent of the project area.	0
g)	Disputes exist in up to 5 percent of the project area.	0
h)	There are no disputes.	0
Q6	What percentage of the project area is affected by disputes regarding access or use rights? Answer: k	
i)	Disputes exist in more than 5 percent of the project area.	0
j)	Disputes exist in up to 5 percent of the project area.	0
k)	There are no disputes.	0
Q7	Have the risks of upstream and marine impacts that could compromise the carbon stocks of a WRC project for the ten years following the risk assessment been demonstrated to be insignificant or effectively mitigated? Answer: m	
l)	Any upstream and marine impacts are not demonstrated to be insignificant or effectively mitigated.	0
m)	Any upstream and marine impacts have been demonstrated to be insignificant or effectively mitigated.	0
Q8	One or more of the following mitigations apply to the project? Answer: No	
n)	Mitigation: The project area is protected by a legally binding agreement (e.g., a licence, a conservation covenant, a conservation easement or a protected area) to continue the management practices that protect the carbon stocks for the entire project crediting period.	0
o)	Mitigation: In the case where disputes exist regarding land possession, ownership, right of access or use, documented evidence is provided demonstrating that the project is taking measures to resolve disputes or clarify overlapping claims.	0
	Total land tenure and resources (LT) = [as applicable, ((a or b) + (c, d, or e) + (f, g or h) + (i, j or k) + (l or m) + n + o)]The Total must not be less than zero	0

	Stakeholder Engagement	Score
Q1	Do local populations, including those living in the project area or within 20 km of the project area boundary, depend on the project area? If No, the risk rating for stakeholder engagement is zero. If the answer is Yes, proceed with Q2. Answer: No	0
Q2	Have more or fewer than 50 percent of stakeholders living in and depending on the project area been consulted?	
a)	Fewer than 50 percent of stakeholders living in and depending on the project area have been consulted.	0
b)	More than 50 percent of stakeholders living in and depending on the project area have been consulted.	0

	Stakeholder Engagement	Score
Q3	Have more or fewer than 20 percent of stakeholders living outside the project area, within 20 km of the project area, and depending on the project area been consulted?	
c)	Fewer than 20 percent of stakeholders living outside the project area, within 20 km of the project area, and depending on the project area have been consulted.	0
d)	More than 20 percent of stakeholders living outside the project area, within 20 km of the project area, and depending on the project area have been consulted.	0
	Total Stakeholder Engagement (SE) = [as applicable, (a or b) + (c or d)]The Total must not be less than zero	0

	Political Risk	Score
Q1	What is the governance score for the Country? Answer: 0.52	
a)	The governance score is below -0.79.	0
b)	Governance score from -0.79 to less than -0.32.	0
c)	Governance score from -0.32 to less than 0.19.	0
d)	Governance score from 0.19 to less than 0.82.	1
e)	Governance score of 0.82 or above.	0
Q2	Does the following mitigation apply to the project? Answer: No	
f)	Mitigation: The project is located in a country that: 1) is party to the Paris Agreement and has submitted a Nationally Determined Contribution (NDC) to the UNFCCC Secretariat in the previous five years; 2) includes AFOLU commitments (conditional or unconditional) in its NDC; and 3) has a documented and active climate change plan that includes the project activity.	0
	Total political risk (PC) [as applicable, ((a, b, c, d or e) + f)]The Total must not be less than zero	1

The governance score was obtained directly from the World Bank portal and was calculated for the period between 2012 and 2022, given that the years 2023 and 2024 are not yet included in the database.

Series Name	2012 [YR2012]	2013 [YR2013]	2014 [YR2014]	2015 [YR2015]	2016 [YR2016]	2017 [YR2017]	2018 [YR2018]	2019 [YR2019]	2020 [YR2020]	2021 [YR2021]	2022 [YR2022]
Control of Corruption: Estimate	0.1720106453	0.07109775394	0.005989762023	0.02710858546	0.08223492652	0.1757457852	0.2159281522	0.2342611849	0.5098643303	0.5174144506	0.5277817249
Government Effectiveness: Estimate	0.4475111663	0.533100903	0.4736343622	0.4913576543	0.5453904271	0.4922479391	0.4090325832	0.4510613382	0.3609854579	0.3260357678	0.4490570724
Political Stability and Absence of Violence/Terrorism: Estimate	0.5082175136	0.4954991937	0.4581654072	0.3632571995	0.3559334278	0.292909652	0.3261395097	0.3810404539	0.3992953598	0.5504139066	0.4135053754
Regulatory Quality: Estimate	0.7445783019	0.7789381146	0.6380656958	0.712536931	0.6978845	0.6911597252	0.7151482701	0.9429237247	0.4877408147	0.5267385244	0.5109314919
Rule of Law: Estimate	0.425940156	0.4363671839	0.4223777652	0.2885879874	0.3675249517	0.319811523	0.2408968806	0.2730046809	0.2077088356	0.2406288087	0.2970094681
Voice and Accountability: Estimate	0.9168092608	0.9542173743	0.9992871284	1.03478992	1.03497839	0.990614295	0.9635645747	0.9051169753	1.054666638	1.084909081	1.070305943
MEAN	0.5358445073	0.5448700873	0.4995866868	0.4862730463	0.5139911038	0.4937481532	0.4784516618	0.5312347263	0.5033769061	0.5410234233	0.5447651794

(Source: WorldWide Global Indicators <https://databank.worldbank.org/source/worldwide-governance-indicators>)

Overall Average: 0.5157423165

Total External Risk
Total External Risk = (LT + SE + PC)
The Total must not be less than zero

9.3 Natural Risks

Natural Risk	Historical Natural Risk Score (LS)	Mitigation (M)	Risk Sub-total (LS×M)	Influenced by Climate Change
a) Fire (F)	4	0.50	2	YES
b) Pest and Disease Epidemics (PD)	30	0.50	15	YES
c) Extreme Climate (W)	5	0.50	2.50	YES
d) Geological Risk (G)*	1	0.50	0.50	NO
e) Other Natural Risks (ON1)	0	0	0	NO
f) Other Natural Risks (ON2)	0	0	0	NO
g) Other Natural Risks (ON3)	0	0	0	NO
h) Sea Level Rise (SLR)	1	0.50	0.50	YES
Expected future impact of climate change on natural risk	Aggregate risk sub-total		Future climate change impact factor	Total (Aggregate risk sub-total × future climate change impact factor)
Natural risk associated with the impact of climate change (NR-c)	Sum of the post-mitigation risk assessment of each historical natural risk influenced by climate change		From 1 to 1.4	20.00
Natural risk NOT associated with the impact of climate change (NR-nc)	Sum of the post-mitigation risk assessment of each historical natural risk not influenced by climate changes		1	0.50
Total Natural Risk = (NR-c + NR-nc + SLR)			Score	
The Total must not be less than zero			20.50	
**If geological risk is landslide indicate "yes" it is influenced by climate change, otherwise indicate "no".				

9.4 Overall Risk Assessment

Having diligently conducted a comprehensive risk assessment using the AFOLU Non-Permanence Risk V4.2 tool, our assessment produced an overall risk score of 39.50 points:

Risk Category	Score
a) Internal Risks	18.00
b) External Risks	1.00
c) Natural Risks	20.50
Overall risk assessment (a+b+c)	39.50

The methodology proposes that if the overall risk assessment exceeds 60, the project risk is considered unacceptably high and the project fails the entire risk analysis. The same applies if any individual element exceeds the following limits:

37. internal risk: 35

38. external risk: 20

39. natural risk: 35

therefore, considering all limits, it is possible to conclude that the risk assessment performed confirms the soundness of the project, not exceeding the limits in any scenario.

10. Agricultural Practices

This section lists the best sustainable and regenerative agricultural practices (BAP) recommended for reducing GHG emissions and increasing carbon sequestration in agriculture. The adoption of these practices can contribute to improving soil health, increasing biodiversity and the resilience of agricultural systems.

10.1: Recommended Agricultural Practices (BAPs)

N o.	Agricultural Practice	Definition
1	Promotion of organic farming (certified and non-certified)	Organic farming is defined by EU Regulation 2018/848.
2a	Zero tillage	Sowing on grassland
2b	Minimum soil tillage	Non-invasive soil tillage to a maximum depth of 15–10 cm
3a	Green cover: spontaneous or sown vegetation	Creation and maintenance of continuous grass cover, which can be either of natural (spontaneous) or intentionally planted (sown) origin.
3b	Use of cover crops	Crops grown to obtain plant biomass incorporated into the soil.
4	Intercropping	The practice of growing two or more crops in a field at the same time.
5	Integrated hedgerows, rows and woods	The establishment of natural or planted hedgerows and windbreaks bordering cultivated land or meadows.

N o.	Agricultural Practice	Definition
6a	Pruning residues as soil amendment	Pruning residues used as amendment.
6b	Pruning residues as energy	Pruning residues used for energy production.
7	Natural substances and leaf-based natural fertilisers	Application of mineral substances such as kaolin and zeolites to soil and leaves.
8	Radical reduction of synthetic fertilisers	Reduction of synthetic fertilisers by at least 15% in the first year.
9	Radical reduction of pesticides	Reduction of pesticides by at least 50% in the first year.
10 a	Organic matter recycling: Agro-industrial waste	Organic waste obtained from the industrial processing of crops (e.g., olive mill waste).
10 b	Organic matter recycling: Biochar	Carbon-rich material obtained from pyrolysis of plant biomass.
10 c	Organic matter recycling: Anaerobic digestate	Semi-liquid OA with fertiliser characteristics obtained from anaerobic digestion.
10 d	Organic matter recycling: Composting	Humus-like material with fertilising characteristics obtained from aerobic digestion of solid waste.
11 a	New plantation: Vine	Conversion from annual cultivation to vineyard plantation.
11 b	New plantation: Orchard	Conversion from annual crop to orchard plantation.
11 c	New plantation: Olive trees	Conversion from annual crop to olive grove plantation.
11 d	New plantation: Other perennial tree species	Conversion from annual crop to other perennial plantation.

10.2: Practices Applied by the Methodology

This section describes in detail the specific BAPs implemented by the methodology. For each practice, the baseline scenario, eligibility conditions and quantification credit table are presented.

10.2.1. Widespread Promotion of Organic Agricultural Management

Description: Compared to conventional agricultural practices, the implementation of organic farming practices has a positive effect on the quantity of CO₂ stored in soil.

Baseline scenario: Conventional crop management (tillage, continuous cropping systems, application of inorganic fertilisers, bare ground fallow between crop rotations).

Eligibility condition: Considered when at least 3 of the following practices are combined: crop rotation, organic fertiliser, maintenance of crop residues and soil cover with green manure crops. Synthetic fertilisers and herbicides are prohibited.

Quantification Table:

Parameter	Estimate (tCO ₂ e/ha/yr)	Source
Soil carbon removal	3.29	Life C-Farms

10.2.2. Conservation Tillage: Minimum Soil Tillage and Zero Tillage

Description: Minimum soil tillage has a dual beneficial effect in reducing CO₂ emissions: it decreases the use of agricultural diesel fuel (derived from fossil fuels) and also increases the quantity of organic matter stored in the soil.

Baseline scenario: Conventional tillage (mouldboard ploughing).

Eligibility condition: This practice is considered only if the use of herbicides is eliminated during pre-sowing and post-harvest phases.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal (Zero tillage)	1.47	Life C-Farms
Soil carbon removal (Minimum tillage)	1.1	Life C-Farms

10.2.3. Grassland / Use of Cover Crops Year-Round

Description: In addition to increasing soil carbon storage, grassland also maintains and increases the level of organic matter in the soil, improving its fertility.

Baseline scenario: Bare soil between crop rotations characterised by the absence of vegetation (application of herbicides or tillage).

Eligibility condition: This practice is considered if no herbicides or soil tillage (of any kind) are used for green cover/spontaneous vegetation. Do not use herbicides as a termination method for cover crop use.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal (Green cover)	2.7	IPCC, Climate Change 2021: The Physical Science Basis
Soil carbon removal (Cover crops)	1.85	Life C-Farms

10.2.4. Intercropping

Description: Intercropping increases agroecosystem biomass by introducing new trees or shrubs. The increase in biomass results in an increase in carbon stocks.

Baseline scenario: Monoculture, i.e. the cultivation of only one crop species in a field at a time.

Eligibility condition: The practice is considered when at least 2 or more crops are grown simultaneously.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Biomass carbon removal	1.01	Life C-Farms

10.2.5. Agricultural Management with Integrated Hedgerows, Rows and Woods in Crops

Description: The presence of buffer strips, hedgerows and/or shrubs to be placed between cultivated land is a practice with innumerable benefits: increased CO₂ absorption, soil health, increased biodiversity, windbreak, aesthetics.

Baseline scenario: Annually cultivated land as land use category.

Eligibility condition: This practice is considered only if the removal of woody vegetation with replanting is considered part of management activities.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Biomass carbon removal	4.0	Life C-Farms

10.2.6. Management of Pruning Residues as Carbon Source for SOC

Description: Even today, pruning residues are often burned. This practice leads to CO₂ emissions into the atmosphere, transforms a resource into waste and can also cause potential accidents and fires. By reusing pruning residues (green and dry).

Baseline scenario: Burning of pruning residues as 'Business as Usual' (BAU) practice.

Eligibility condition: Biomass combustion not associated with energy production is not permitted. This practice is considered only when plant biomass is not burned (either on site or within the boundaries of the entire farm) in order to consider the phenomenon of dispersion.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal (Residues as amendment)	2.7	Life C-Farms
CO ₂ emissions avoided from combustion (Residues as energy)	1.85	Life C-Farms

10.2.7. Application of Natural Inorganic Substances and Natural Fertilisers

Description: This practice focuses on replacing or reducing the use of synthetic fertilisers with natural, low environmental impact alternatives, such as natural inorganic substances and leaf-based natural fertilisers (rocks or mineral dusts).

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal	0.74	Life C-Farms

10.2.8. Radical Reduction of Synthetic Fertilisers

Description: The reduction in the use of chemical fertilisers makes it possible to avoid the release of nitrous oxide into the atmosphere, a powerful greenhouse gas.

Eligibility condition: This practice is considered if no synthetic fertilisers are used or if their progressive elimination is underway.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
N ₂ O emissions avoided	1.27	Life C-Farms

10.2.9. Radical Reduction of Pesticides

Description: The intensive use of pesticides kills the soil biome and leads to a drastic reduction in the soil's carbon storage capacity. Pesticides can also release greenhouse gases into the atmosphere, such as nitrous oxide, when applied to crops.

Eligibility condition: This practice is considered if no herbicides are used or if their progressive elimination is underway.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal	0.28	Life C-Farms

10.2.10. Optimal Organic Matter Recycling

Description: 'Residual' organic matter (such as plant material and animal manure) when it decomposes can release greenhouse gases into the atmosphere (e.g., methane). Optimal recycling of organic matter, such as the use of farm-produced biomass, can potentially generate saleable carbon credits from both emission reduction and improvement of soil fixation.

Baseline scenario: Application of inorganic nitrogen fertilisers (INF).

Eligibility condition: This practice is considered only when the plant biomass from which the organic amendment (OA) is derived has been grown on the same farm where it is applied. Alternatively, purchased OA applied to agricultural land may be considered eligible if produced within regional boundaries or within a radius of 5–100 km and if the OA seller/producer does not benefit from certified carbon removals. The application of OA is considered eligible only for an equivalent nitrogen dose.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Soil carbon removal (Agro-industrial waste)	2.5	Life C-Farms
Soil carbon removal (Biochar)	2.5	Life C-Farms
Soil carbon removal (Anaerobic digestate)	2.5	Life C-Farms
Soil carbon removal (Composting)	2.5	Life C-Farms

10.2.11. New Plantations

Description: The activity consists of establishing a new olive grove (or orchard) on abandoned, unused or previously cultivated or grazed land. This would allow greater absorption of CO₂ from the atmosphere.

Baseline scenario: Conversion from annual cultivation to perennial tree plantation.

Eligibility condition: The conversion or maintenance of perennial tree crops will be considered eligible when: permanent soil cover (planted or spontaneous) is maintained. This practice is not mandatory during summer. The mouldboard plough is replaced with a reduced soil disturbance technique.

Parameter	Estimate (t CO ₂ /ha/yr)	Source
Biomass carbon removal (Vine)	1.8	Life C-Farms
Biomass carbon removal (Orchard)	2.6	Life C-Farms
Biomass carbon removal (Olive grove)	2.2	Life C-Farms
Biomass carbon removal (Other species)	1.5	IPCC, Climate Change 2021: The Physical Science Basis

11. Parameters, Formulas and Verification

This section details the quantification parameters and formulas used for calculating and verifying GHG emissions and removals associated with BAPs, both before (baseline) and after implementation of the practice.

11.1: Recommended Agricultural Practices (BAPs)

1. Organic Farming (Certified and Non-Certified)

Description: Promotes the adoption of organic farming practices, as defined by EU Regulation 2018/848, to increase the quantity of CO₂ stored in soil.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and conventional agricultural practices in reference area. Soil sampling and analysis to determine organic carbon content (SOC) under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in organically managed soils. Comparison with pre-implementation data to assess SOC increment attributable to the practice.
Soil carbon removal (post-implementation)	3.29	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data in organically managed soils. Comparison with pre-implementation data and reference values for

Parameter	Estimate (tCO ₂ e/h a/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
				similar organic systems to assess the effectiveness of the practice.

2a. Zero Tillage

Description: Eliminates soil tillage, seeding directly onto grassland.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/h a/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and conventional tillage practices in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in zero-tillage managed soils. Comparison with pre-implementation data to assess SOC increment attributable to the practice.
Soil carbon removal (post-implementation)	1.47	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data in zero-tillage managed soils. Comparison with pre-implementation data and reference values for zero-tillage systems.

2b. Minimum Soil Tillage

Description: Reduces the intensity of soil tillage, limiting depth to 15–10 cm.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/h a/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and conventional tillage practices in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in minimum-tillage managed soils. Comparison with pre-implementation data.

Parameter	Estimate (tCO ₂ e/h a/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (post-implementation)	1.1	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data in minimum-tillage managed soils. Comparison with pre-implementation data and reference values for minimum-tillage systems.

3a. Green Cover: Spontaneous or Sown Vegetation

Description: Maintains continuous grass cover, both natural and sown, to protect and improve soil.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/h a/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and soil management practices without green cover in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in green-cover managed soils. Comparison with pre-implementation data.
Soil carbon removal (post-implementation)	2.7	Soil carbon models (e.g. RothC, Century) used to simulate carbon dynamics based on management practices.	N/A	Analysis of SOC monitoring data in green-cover managed soils. Comparison with pre-implementation data, soil carbon model results and reference values to assess the effectiveness of the practice.

3b. Use of Cover Crops

Description: Grows specific crops to obtain plant biomass to be incorporated into soil as mulch.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and soil management practices without cover crops in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in cover-crop managed soils. Comparison with pre-implementation data.
Soil carbon removal (post-implementation)	1.85	Soil carbon models (e.g. RothC, Century) used to simulate carbon dynamics based on management practices.	N/A	Analysis of SOC monitoring data in cover-crop managed soils. Comparison with pre-implementation data, soil carbon model results and reference values to assess effectiveness.

4. Intercropping

Description: Simultaneously grows two or more crops in the same field to increase biomass and diversity.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Biomass carbon removal (baseline)	0	N/A, default value	Analysis of historical data and monoculture practices in reference area. Estimation of biomass produced under monoculture conditions and its carbon content.	Biomass sampling and analysis in intercropping systems, to determine total biomass and carbon content. Comparison with pre-implementation data.
Biomass carbon removal (post-implementation)	1.01	Species-specific growth models to estimate biomass accumulation and carbon sequestration.	N/A	Analysis of biomass and carbon sequestration monitoring data in intercropping systems. Comparison with growth model results and reference values to assess effectiveness.

5. Integrated Hedgerows, Rows and Woods

Description: Integrates hedgerows, rows and woods into cultivated fields to increase carbon sequestration, improve soil health and biodiversity.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Biomass carbon removal (baseline)	0	N/A, default value	Analysis of historical data and soil management practices without hedgerows, rows or woods in reference area. Estimation of woody biomass present under BAU conditions and its carbon content.	Periodic sampling and measurement of trees (breast height diameter (DBH), height) in hedgerows, rows and woods, to estimate woody biomass and carbon content. Comparison with pre-implementation data.
Biomass carbon removal (post-implementation)	4.0	Species-specific allometric models to estimate biomass and carbon removal as a function of DBH and height.	N/A	Analysis of woody biomass and carbon sequestration monitoring data. Comparison with allometric model results and reference values for similar agroforestry systems to assess effectiveness.

6a. Pruning Residues as Amendment

Description: Uses pruning residues as soil amendment to increase organic matter content.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and pruning residue management practices in reference area (e.g. burning). Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in soils where pruning residues are used as amendment. Comparison with pre-implementation data.
Soil carbon removal (post-implementation)	2.7	Organic matter decomposition models (e.g. RothC, Century)	N/A	Analysis of SOC monitoring data. Comparison with pre-implementation

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
		to estimate carbon sequestered following incorporation.		data, model results and reference values to assess effectiveness.

6b. Pruning Residues as Energy

Description: Uses pruning residues for energy production, replacing fossil fuels.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
CO ₂ emissions avoided from combustion (baseline)	*	Quantity of burned residues (t/ha) × Combustion emission factor (t CO ₂ /t residues)	Quantify the quantity of pruning residues burned under BAU conditions, using historical data and management practice information.	Monitor the quantity of pruning residues used for energy production and verify the corresponding reduction in fossil fuel use.
CO ₂ emissions avoided from combustion (post-implementation)	1.85	Quantity of managed residues (t/ha) × Combustion emission factor (t CO ₂ /t residues)	N/A	Calculation of CO ₂ emissions avoided based on quantity of residues used and emission factor. Comparison with pre-implementation data.

* Value estimated based on quantity of burned residues, which varies depending on crop type and management practices.

7. Application of Natural Inorganic Substances and Leaf-Based Natural Fertilisers (Rocks or Mineral Dusts)

Description: This practice focuses on replacing or reducing the use of synthetic fertilisers with natural, low environmental impact alternatives, such as natural inorganic substances and leaf-based natural fertilisers (rocks or mineral dusts).

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and conventional fertilisation	Periodic soil sampling and analysis to monitor changes in SOC content in soils

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
			practices in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	managed with the application of natural substances and fertilisers. Comparison with pre-implementation data.
Soil carbon removal (post-implementation)	0.74	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data in soils managed with the application of natural substances and fertilisers. Comparison with pre-implementation data and reference values for similar systems to assess effectiveness.

8. Radical Reduction of Synthetic Fertilisers

Description: The reduction in the use of chemical fertilisers makes it possible to avoid the release of nitrous oxide into the atmosphere, a powerful greenhouse gas.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
N ₂ O emissions (baseline)	*	Quantity of nitrogen fertiliser applied (kg N/ha) × EF (kg CO ₂ e/kg N)	Quantify the quantity of nitrogen fertiliser applied under BAU conditions, using historical data and management practice information. Determine the appropriate emission factor (EF).	Monitor the quantity of nitrogen fertiliser applied after the reduction and calculate N ₂ O emissions. Comparison with pre-implementation data to assess the reduction in emissions.
N ₂ O emissions (post-implementation)	**	Quantity of nitrogen fertiliser applied after reduction (kg N/ha) × EF (kg CO ₂ e/kg N)	N/A	Calculation of N ₂ O emissions based on the quantity of fertiliser applied and the emission factor. Comparison with pre-implementation data and reference values for systems with reduced fertilisation.

* Value estimated based on quantity of nitrogen fertiliser applied, which varies depending on crop type and management practices.

** Value estimated based on quantity of nitrogen fertiliser applied after reduction, which should be lower than BAU.

9. Radical Reduction of Pesticides

Description: Reduces the use of pesticides to protect the soil biome and its carbon sequestration capacity.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (baseline)	0	N/A, default value	Analysis of historical data and soil management practices with intensive pesticide use in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in soils managed with reduced pesticide application. Comparison with pre-implementation data.
Soil carbon removal (post-implementation)	0.28	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data in soils managed with reduced pesticide application. Comparison with pre-implementation data and reference values for similar systems to assess effectiveness.

10. Optimal Organic Matter Recycling

Description: 'Residual' organic matter (such as plant material and animal manure) when it decomposes can release greenhouse gases into the atmosphere (e.g., methane). Optimal recycling of organic matter, such as the use of farm-produced biomass, can potentially generate saleable carbon credits from both emission reduction and improvement of soil fixation.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Soil carbon removal (Agro-industrial waste - baseline)	0	N/A, default value	Analysis of historical data and agro-industrial waste management	Periodic soil sampling and analysis to monitor changes in SOC content in soils where agro-industrial waste is

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
			practices (e.g. disposal) in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	used as amendment. Comparison with pre-implementation data.
Soil carbon removal (Agro-industrial waste - post-implementation)	2.5	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data. Comparison with pre-implementation data, model results and reference values to assess effectiveness.
Soil carbon removal (Biochar - baseline)	0	N/A, default value	Analysis of historical data and biochar management practices (e.g. non-use) in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in soils where biochar is used as amendment. Comparison with pre-implementation data.
Soil carbon removal (Biochar - post-implementation)	2.5	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data. Comparison with pre-implementation data, model results and reference values to assess effectiveness.
Soil carbon removal (Anaerobic digestate - baseline)	0	N/A, default value	Analysis of historical data and anaerobic digestate management practices (e.g. non-use) in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in soils where anaerobic digestate is used as amendment. Comparison with pre-implementation data.
Soil carbon removal (Anaerobic digestate - post-implementation)	2.5	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data. Comparison with pre-implementation data, model results and reference values to assess effectiveness.

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
post-implementation)				reference values to assess effectiveness.
Soil carbon removal (Composting - baseline)	0	N/A, default value	Analysis of historical data and compost management practices (e.g. non-use) in reference area. Soil sampling and analysis to determine SOC content under BAU conditions.	Periodic soil sampling and analysis to monitor changes in SOC content in soils where compost is used as amendment. Comparison with pre-implementation data.
Soil carbon removal (Composting - post-implementation)	2.5	N/A, default value derived from comparative studies	N/A	Analysis of SOC monitoring data. Comparison with pre-implementation data, model results and reference values to assess effectiveness.

11. New Plantations

Description: The activity consists of establishing a new olive grove (or orchard) on abandoned, unused or previously cultivated or grazed land. This would allow greater absorption of CO₂ from the atmosphere.

Quantification parameters:

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
Biomass carbon removal (Vine - baseline)	0	N/A, default value	Analysis of historical data and land use in reference area (e.g. annual crops). Estimation of woody biomass present under BAU conditions and its carbon content (may be negligible).	Periodic sampling and measurement of vines (DBH, height) to estimate woody biomass and carbon content. Comparison with pre-implementation data.
Biomass carbon removal (Vine - post-implementation)	1.8	Species-specific allometric models for vine, to estimate biomass and carbon removal as a	N/A	Analysis of woody biomass and sequestered carbon monitoring data. Comparison with allometric model results and reference values for

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
		function of DBH and height.		similar systems to assess effectiveness.
Biomass carbon removal (Orchard - baseline)	0	N/A, default value	Analysis of historical data and land use in reference area (e.g. annual crops). Estimation of woody biomass present under BAU conditions and its carbon content (may be negligible).	Periodic sampling and measurement of fruit trees (DBH, height) to estimate woody biomass and carbon content. Comparison with pre-implementation data.
Biomass carbon removal (Orchard - post-implementation)	2.6	Species-specific allometric models for orchard tree species, to estimate biomass and carbon removal as a function of DBH and height.	N/A	Analysis of woody biomass and sequestered carbon monitoring data. Comparison with allometric model results and reference values for similar systems to assess effectiveness.
Biomass carbon removal (Olive grove - baseline)	0	N/A, default value	Analysis of historical data and land use in reference area (e.g. annual crops). Estimation of woody biomass present under BAU conditions and its carbon content (may be negligible).	Periodic sampling and measurement of olive trees (DBH, height) to estimate woody biomass and carbon content. Comparison with pre-implementation data.
Biomass carbon removal (Olive grove - post-implementation)	2.2	Species-specific allometric models for olive trees, to estimate biomass and carbon removal as a function of DBH and height.	N/A	Analysis of woody biomass and sequestered carbon monitoring data. Comparison with allometric model results and reference values for similar systems to assess effectiveness.
Biomass carbon removal (Other species - baseline)	0	N/A, default value	Analysis of historical data and land use in reference area (e.g. annual crops). Estimation of woody biomass present under	Periodic sampling and measurement of trees of other species (DBH, height) to estimate woody biomass and carbon content. Comparison with

Parameter	Estimate (tCO ₂ e/ha/yr)	Formula	Verification (Pre-implementation)	Verification (Post-implementation)
			BAU conditions and its carbon content (may be negligible).	pre-implementation data.
Biomass carbon removal (Other species - post-implementation)	1.5	Species-specific allometric models for the tree species used, to estimate biomass and carbon removal as a function of DBH and height.	N/A	Analysis of woody biomass and sequestered carbon monitoring data. Comparison with allometric model results and reference values for similar systems to assess effectiveness.

12. Glossary of Terms and Acronyms

Terms

Additionality: Principle establishing that GHG reductions/removals generated by a project or activity are additional to those that would have occurred in the absence of the intervention.

Organic farming: Agricultural production system that excludes the use of synthetic inputs such as fertilisers and pesticides, promoting biodiversity, biological cycles and soil health.

Conservation tillage: Soil tillage techniques that minimise soil disturbance, such as zero tillage and minimum tillage.

Baseline (Reference scenario): Level of GHG emissions and/or removals that would occur in the absence of a mitigation project or activity.

Biochar: Carbon-rich material obtained from pyrolysis of biomass, which can be used as a soil amendment to increase fertility and sequester carbon.

Biomass: Organic material derived from plants and animals.

Intercropping: Agricultural practice involving the cultivation of two or more crops in the same field simultaneously.

Composting: Process of controlled aerobic decomposition of organic waste, which produces a humus-like material usable as an amendment.

Green cover: Maintenance of plant cover on the soil, both natural and sown, to protect and improve the soil.

Carbon credits: Certificates representing one tonne of CO₂ equivalent removed or reduced from the atmosphere, generated by GHG mitigation projects.

Anaerobic digestate: By-product of biogas production, rich in nutrients and usable as an organic fertiliser.

GHG emissions: Release of greenhouse gases into the atmosphere, resulting from human or natural activities.

Emission factors: Coefficients that quantify GHG emissions associated with specific activities or processes.

Synthetic fertilisers: Industrially produced fertilisers, containing nutrients in chemical form.

Greenhouse gases (GHG): Gases present in the atmosphere that trap heat, contributing to global warming.

GHG inventory: Accounting of GHG emissions and/or removals of an organisation, project or product.

Minimum soil tillage: Soil cultivation technique that minimises soil disturbance, limiting tillage depth.

Best Agricultural Practices (BAP): Agricultural management techniques that promote environmental, economic and social sustainability.

Monoculture: Cropping system involving the cultivation of only one plant species in a field at a time.

Notarisation: Process that ensures the authenticity, integrity and traceability of carbon credits, using secure and transparent digital registers.

Pesticides: Chemical substances used to control organisms harmful to crops.

Net GHG reduction: Occurs when an agricultural or soil management activity reduces GHG emissions relative to the baseline scenario (BAU). This means that the intervention adopted has led to lower emissions of CO₂ or other greenhouse gases. Examples: transition from conventional to minimum tillage to reduce CO₂ emissions from soil organic matter oxidation; replacing chemical fertilisers with organic amendments, reducing nitrous oxide (N₂O) emissions; optimising livestock waste management to reduce methane (CH₄) production.

Net GHG removal: Occurs when an agricultural or agroforestry system absorbs more greenhouse gases than the baseline, increasing CO₂ sequestration in the environment. This occurs through biological processes that trap carbon, such as photosynthesis and accumulation of organic matter in soil. Examples: creation of hedgerows and tree strips around cultivated fields, which sequester CO₂ through biomass growth; adoption of cover crops (cover crops) to increase soil carbon content; implementation of agroforestry systems, combining crops and trees to improve CO₂ removal from the atmosphere.

Crop rotation: Agricultural practice involving the alternation of different crops on the same land over the years.

Carbon sequestration: Process of capturing and storing atmospheric carbon in the long term, e.g. in soil or biomass.

Integrated hedgerows, rows and woods: Integration of tree and shrub elements in agricultural systems, to provide various ecosystem services, including carbon sequestration.

GHG sources, sinks and reservoirs (SSR): Elements that contribute to GHG emissions and/or removals. Sources release GHGs, sinks remove them, reservoirs accumulate them.

Natural inorganic substances: Substances of mineral origin usable as amendments or fertilisers, which have not undergone chemical synthesis processes.

Acronyms

AFOLU: Agriculture, Forestry and Other Land Use

ALM: Agricultural Land Management

ARR: Afforestation, Reforestation and Revegetation

BAU: Business As Usual

BAP: Best Agricultural Practices

CO₂: Carbon dioxide, also known as carbonic anhydride

CDM: Clean Development Mechanism

DAP: Diameter at Breast Height

GHG: Greenhouse Gas

IPCC: Intergovernmental Panel on Climate Change

N₂O: Nitrous oxide (also known as dinitrogen oxide or laughing gas)

OA: Organic Amendment

SOC: Soil Organic Carbon

VCS: Verified Carbon Standard